Amplitude-Robust Qubit Spectroscopy Using Echo-Root-Lorentzian Pulse Shapes

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We present a spectroscopy method for driven two-level systems based on a root-Lorentzian envelope with a π-phase inversion at its peak. For driven-spectroscopy pulses shorter than the lifetime T_1, the resulting echo-like symmetry cancels on-resonant Rabi oscillations while preserving the off-resonant spectral response and suppressing conventional power broadening. Measurements on a superconducting circuit show a narrow echo feature over more than three decades of peak drive, extending the amplitude range over which near-T_2-limited spectroscopy can be performed. Across peak Rabi frequencies up to 50 MHz, the measured response remains centered near resonance without the monotonic AC-Stark displacement expected for a constant drive. To suppress the remaining center motion, we derive an accumulated-phase AC-Stark compensation protocol and experimentally reduce approximately 10 kHz resonance-center excursions to below 1 kHz. A direct comparison of measured spectra with numerical calculations validates the model and supports a practical approach to amplitude-robust, high-resolution spectroscopy.

Determining the transition frequency of a quantum system is a fundamental task in spectroscopy, quantum metrology, and quantum sensing. The simplest setting is a two-level system, whose energy splitting defines a spectroscopic resonance and can encode changes in its local environment. This frequency is commonly estimated by sweeping a coherent drive across the transition and monitoring the resulting population response [1–4]. Such measurements underlie both the characterization of quantum devices and the use of quantum systems as probes of external perturbations.

In direct population spectroscopy, spectral resolution and measurement robustness are competing requirements. Conventional continuous-wave spectroscopy generally applies a drive long enough for the population to approach a steady state. In the weak-drive regime, decoherence produces a homogeneous linewidth that scales as $1/T_2$, establishing the natural coherence-limited resolution scale of the transition. Increasing the drive strength makes the population response more pronounced, but also power broadens the transition. Conventional high-resolution spectroscopy therefore requires the drive amplitude to be chosen carefully, balancing a pronounced population response against power broadening. High-amplitude spectroscopy is further complicated when the nominal two-level system is embedded in a multilevel spectrum. Off-resonant coupling to neighboring transitions can produce amplitude-dependent AC-Stark shifts, line-shape distortions, and leakage, causing the apparent resonance to move with drive strength [5–7].

Ramsey interferometry provides a standard alternative to direct spectroscopy and, in principle, enables frequency estimation at the T_2 limit [2]. Other sensing protocols can surpass this probe-T_2 scale using inter-shot correlations, external references, dynamical decoupling, or tailored filter functions [3, 4, 8–11]. Ramsey generally presupposes an approximate transition frequency and calibrated $\pi/2$ pulses for state preparation and readout, and is therefore not fully calibration independent. Achieving T_2-limited resolution also requires long interrogation times and sufficiently dense temporal sampling to resolve the resulting interference fringes. Finite-pulse imperfections and probe-induced frequency shifts, including unintended AC-Stark shifts during the control pulses, can nevertheless bias the extracted transition frequency [12, 13].

Here, we propose a spectroscopic method for probing the qubit transition frequency using an echo-root-Lorentzian drive pulse. Previous theoretical work showed that Lorentzian pulses exhibit power narrowing, complementary to the conventional power broadening observed for constant drives, as a consequence of their adiabatic properties [14]. Power narrowing with powers-of-Lorentzian pulses was subsequently demonstrated experimentally on IBM Quantum, including the cutoff broadening introduced by truncating the pulse wings [15]. In this work, we extend this framework to longer pulse durations approaching the T_1 timescale and introduce a π-phase jump at the pulse peak. The resulting echo property effectively cancels the net pulse area while preserving the adiabatic properties of the drive. This phase inversion reverses the coherent phase accumulation at resonance, suppressing on-resonant excitation while maintaining an off-resonant spectral response across the frequency domain, and thus making resonance distinguishable.

We define the echo-root-Lorentzian drive from the order-n generalized Lorentzian envelope as

$$L^{(n)}(t) = \frac{1}{\left[1 + \left(\frac{t}{\tau}\right)^2\right]^n}, \tag{1}$$

$$L^{(1/2)}_{\text{echo}}(t) := L^{(1/2)}(t)\,\mathrm{sgn}(t). \tag{2}$$

where τ sets the characteristic pulse duration and n controls the algebraic decay of the wings.

For the untruncated, plain generalized Lorentzian with $n > 1/2$, the boundary of adiabatic following provides an

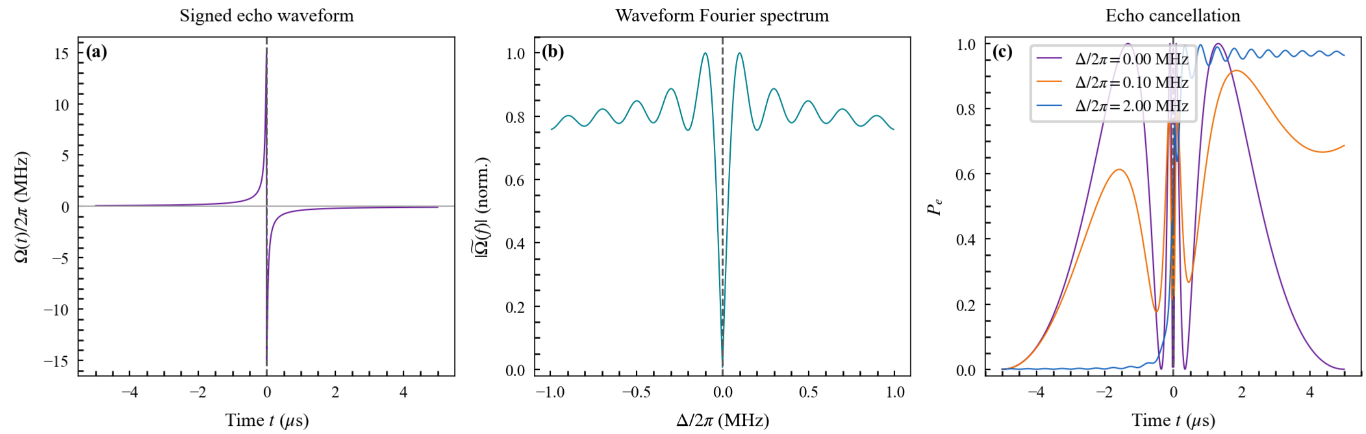

Figure 1. Pulse waveform and echo-cancellation mechanism. (a) Signed echo-root-Lorentzian drive envelope with peak amplitude $\Omega_0/(2\pi) = 15$ MHz and a π phase inversion at the pulse midpoint. (b) Normalized magnitude of the waveform Fourier spectrum, showing the spectral null at zero frequency produced by the phase inversion. (c) Unitary two-level excited-state population during the pulse at $\Delta/(2\pi) = 0$, 0.10, and 2.00 MHz, without relaxation or dephasing. At resonance, the evolution accumulated during the first half is exactly reversed during the second half, whereas finite detuning prevents complete cancellation.

estimate of the characteristic spectral width. Applying the usual adiabatic condition to this envelope gives [14]

$$\Delta_b \tau \propto (\Omega_0 \tau)^{-\frac{1}{2n-1}}, \tag{3}$$

where Δ_b is the angular detuning at which adiabatic following ceases and Ω_0 is the peak angular Rabi frequency. Because the exponent is negative, this characteristic detuning decreases as the drive strength increases, producing power narrowing rather than conventional power broadening. The exponent grows without bound as $n \to 1/2^+$, so within its domain this asymptotic result predicts the steepest narrowing trend at the root-Lorentzian boundary. At $n = 1/2$, finite truncation, decoherence, and the echo phase inversion regularize the response, and the FWHM Γ_f used below is therefore obtained from the complete experimental spectrum. This interpretation is consistent with the power narrowing and cutoff broadening observed for powers of Lorentzian pulses by Mihov and Vitanov [15]. In contrast, a square pulse exhibits conventional power broadening, with $\Gamma_\omega \propto \Omega_0$ at large drive. The derivation and the marginal $n = 1/2$ limit are detailed in the Supplemental Material [16].

Square pulses exhibit pronounced power broadening even at low drive amplitudes compared to Lorentzian-type pulses, with substantial linewidth expansion observed over three orders of magnitude in amplitude. In contrast, the root-Lorentzian pulse displays the characteristic power-narrowing behavior and maintains a narrow linewidth. However, even for pulse durations shorter than the qubit lifetime T_1, coherent oscillations arise for both square and root-Lorentzian pulses, resulting in amplitude-dependent linewidth variations that hinder precise spectral extraction.

Introducing a π phase inversion at the peak of the root-Lorentzian envelope partitions the drive into two sequential segments of opposite phase. For an ideal symmetric unitary two-level pulse, the corresponding forward and backward evolutions cancel exactly at zero detuning; relaxation, dephasing, waveform imbalance, and finite bandwidth make the cancellation incomplete in experiment. This echo-like structure suppresses resonant Rabi oscillations while preserving the off-resonant spectral response, producing a sharp depletion feature centered at the qubit transition frequency. The waveform Fourier spectrum in Fig. 1 has a null at zero baseband frequency, consistent with its vanishing signed area, but this linear waveform property is not itself the nonlinear qubit response. The resonance-centered depletion instead follows from the driven dynamics and is observed in the experimental spectroscopy maps.

This mechanism can be viewed as a Loschmidt-echo-like evolution, in which ideal reversibility occurs only on resonance, while detuning, decoherence, and experimental imperfections break the cancellation and restore excitation [17, 18]. Combined with the adiabatic following of the Lorentzian envelope [14, 15], this interference effect suppresses both power broadening and coherent Rabi oscillations, enabling narrow spectroscopy over a broad range of drive strengths with driven-spectroscopy pulses shorter than T_1. This sub-T_1 operating regime remains useful as qubit lifetimes increase because square and ordinary root-Lorentzian pulses can still exhibit coherent oscillatory distortions. The numerical model used for the experiment–theory comparison in Fig. 3 is detailed in the Supplemental Material [16].

Fig. 2 compares measured spectroscopy for a constant pulse, an ordinary root-Lorentzian pulse, and an echo-

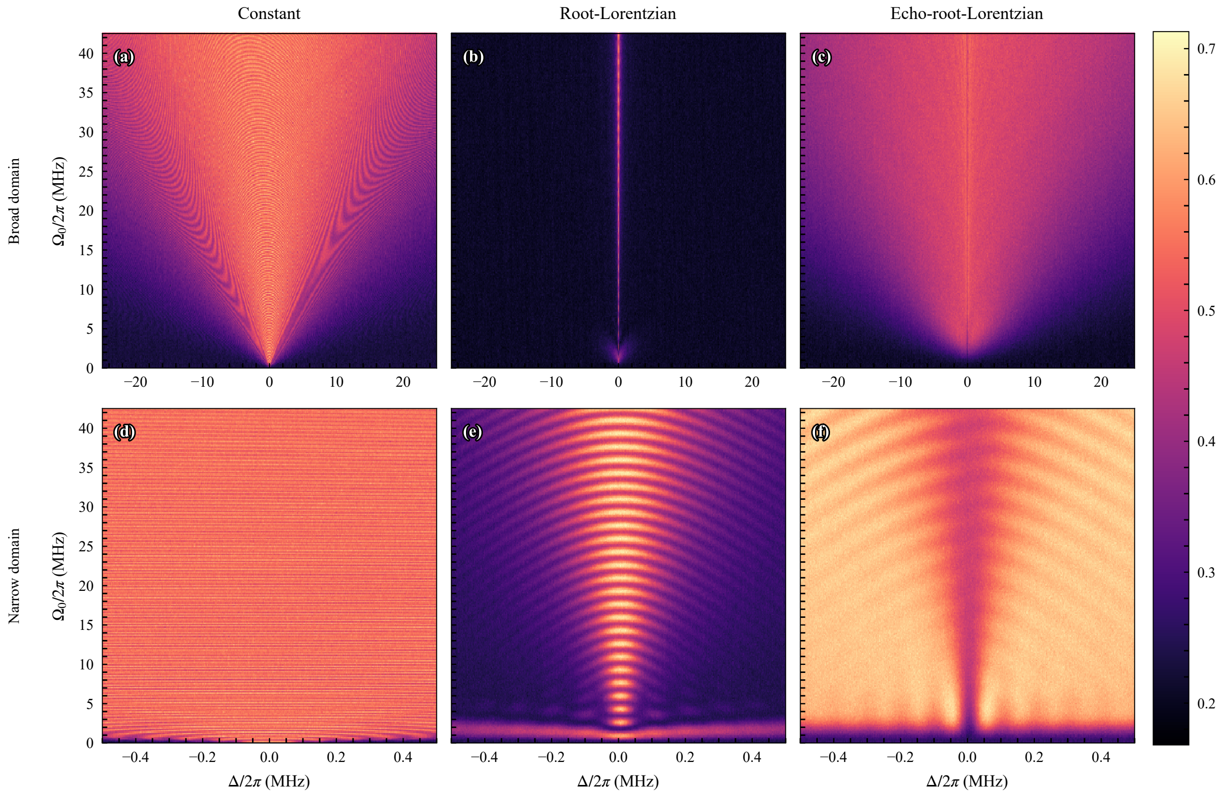

Figure 2. Experimental q1 spectroscopy maps for 20 μs pulses. The color scale is the state-discriminated, measured excited-state population P_e. The upper row shows the broad 50 MHz domain (-25 to $+25$ MHz); the lower row shows the narrow 1 MHz domain (-0.5 to $+0.5$ MHz). From left to right, the columns show a nearly constant pulse implemented as a root-Lorentzian with $c = 0.999$ and no phase inversion, an ordinary root-Lorentzian with $c = 0.005$, and an echo-root-Lorentzian with $c = 0.005$. The calibrated peak-Rabi range extends from zero to approximately 42.5 MHz. Each point uses 2000 shots; active reset and state-discriminated readout are used throughout.

root-Lorentzian pulse on q1. All protocols use $L = 20$ μs and a calibrated peak-Rabi range extending to approximately 42.5 MHz. The upper and lower rows span broad 50 MHz and narrow 1 MHz detuning domains, respectively. The constant-envelope and ordinary root-Lorentzian measurements retain strong amplitude-dependent coherent structure. The echo-root-Lorentzian data instead show a narrow resonance-centered depletion channel over the measured drive range.

Fig. 3(a)–(f) provides a complementary fixed-amplitude comparison for the ordinary and echo protocols. These traces use $L = 20$ μs and $c = 0.005$. The three rows compare the protocols at $\Omega_0/(2\pi) = 3$, 20, and 40 MHz, making the echo-induced central depletion and its persistence with increasing drive amplitude explicit.

To identify the useful operating regime, we examine the linewidth and contrast of the echo response as functions of pulse duration and drive amplitude. We choose pulse durations shorter than T_1, but long enough that Fourier broadening does not dominate. The duration-dependent linewidth, contrast, and contrast-weighted resolution are reported in the Supplemental Material [16].

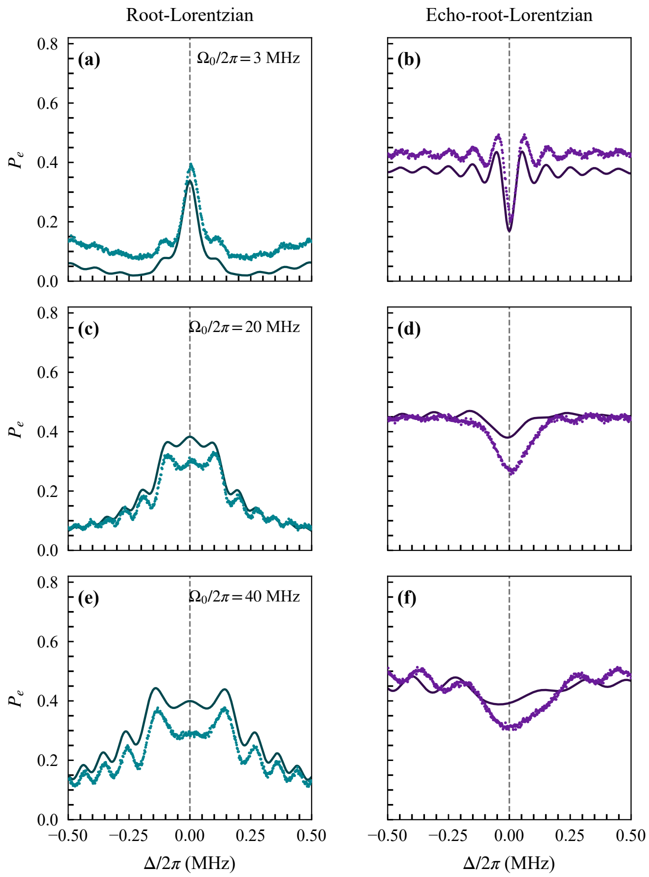

Figure 3. Experimental and simulated spectra. (a)–(f) Final excited-state probability P_e versus detuning for 20 μs root-Lorentzian (left) and echo-root-Lorentzian (right) pulses with $c = 0.005$. From top to bottom, $\Omega_0/2\pi = 3$, 20, and 40 MHz. Colored continuous lines show three-level simulations evaluated with 8000 temporal steps per pulse half, and colored markers show the 30 August 2026 q6 OPX1000 data acquired with 10,000 shots per detuning point. For lineshape comparison, the same empirical display map $P_e^{\mathrm{plot}} = 1.02945 P_e^{\mathrm{raw}} - 0.19488$ is applied to all experimental points; it is not a readout-assignment correction.

A second design constraint is the cutoff of the formally infinite-support pulse envelope. Truncation introduces a residual endpoint drive $\Omega_c = c\Omega_0$ and a corresponding cutoff-broadening contribution [15, 19, 20]. The relevant weak-saturation condition is $s_c = \Omega_c^2 T_1 T_2 \ll 1$, or $\Omega_c \ll 1/\sqrt{T_1 T_2}$; the inequality therefore points toward *smaller*, not larger, endpoint amplitudes. The relative cutoff c cannot be specified independently of Ω_0, T_1, and T_2, giving a three-dimensional design space of pulse duration, drive amplitude, and cutoff level. The corresponding cutoff–amplitude measurements and their resolution–signal analysis are reported in the Supplemental Material [16].

Beyond linewidth and contrast, amplitude robustness requires that increasing the drive not displace the inferred resonance center. At large amplitudes, multilevel AC-Stark shifts and finite-pulse interference become important; we therefore examine the center stability over the same 50 MHz peak Rabi-frequency span used in Fig. 2.

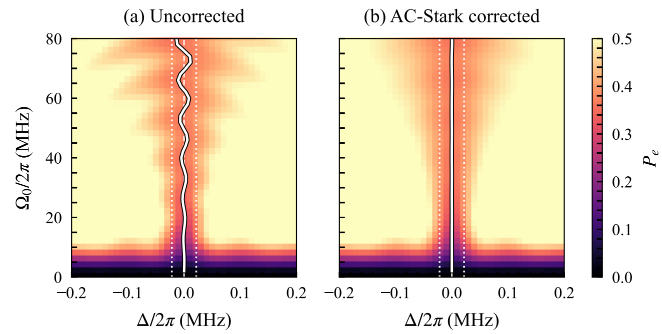

Figure 4. Echo-root-Lorentzian spectroscopy without (a) and with (b) accumulated-phase compensation of drive-induced center motion. The maps show the measured excited-state population for $L = 20$ μs, cutoff $c = 0.001$, and $\beta = 0$; the applied correction is $\kappa = 0.00225$ MHz^{-1}. White curves mark the fitted spectral centers. The dashed line marks zero detuning, and the paired dotted lines at $\Delta/(2\pi) = \pm\Gamma_{f,T_2}/2$ enclose the coherence-limited FWHM $\Gamma_{f,T_2} = 1/(\pi T_{2,\mathrm{eff}}) = 43.5$ kHz. The common color scale is clipped at $P_e = 0.5$ for visibility. The uncorrected center oscillates with drive amplitude, whereas the corrected center remains pinned near zero detuning.

For comparison, the weak-drive three-level AC-Stark shift of a constant drive is $\delta f_{01} \simeq -(\Omega_0/2\pi)^2/[2(\alpha/2\pi)]$ [5, 21]. With $\alpha/(2\pi) = -216.0$ MHz, it reaches 5.8 MHz at $\Omega_0/(2\pi) = 50$ MHz. For the 20 μs root-Lorentzian envelope with $c = 0.001$, the phase-average shift is smaller by $\langle f^2 \rangle_t = 0.001570$ and reaches only 9.1 kHz; the echo sign change does not cancel this quadratic term. Fig. 4 shows the corresponding two-dimensional echo response before and after applying the accumulated-phase correction. Without correction, the operational spectral center oscillates with pulse area; after correction, it remains locked near zero detuning. Fig. 5 displays the measured scale separation through 60 MHz peak Rabi frequency. The full response maps and fine central-frequency scans are given in the Supplemental Material [16]. The corrected response shows none of the approximately quadratic displacement expected for a constant drive.

In conclusion, we have demonstrated a narrow spectroscopic depletion feature that persists over more than three decades of peak drive. Measurements establish its linewidth, contrast, drive-induced center motion, and accumulated-phase compensation. The agreement between the measured spectra and the numerical curves in Fig. 3 validates the model. Phase-inverted shaped pulses therefore provide a practical route to high-resolution qubit spectroscopy beyond the weak-drive regime.

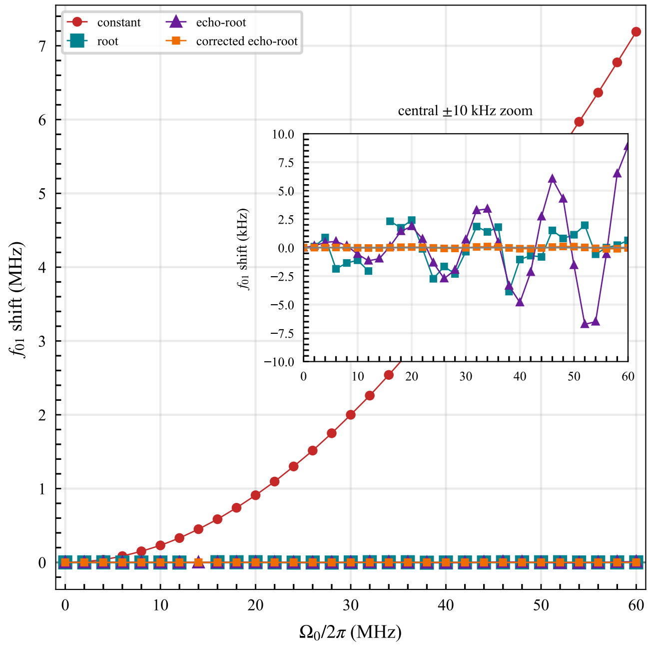

Figure 5. Measured drive-induced f_{01} displacement over an extended peak-Rabi-frequency range. The curves connect the measured spectral centers for the constant-pulse and root-Lorentzian references and for the uncorrected and accumulated-phase-corrected echo-root-Lorentzian response. The root reference uses $L = 10\ \mu$s and $c = 0.002$; both echo curves use $L = 20\ \mu$s, $c = 0.001$, $\beta = 0$, and $\alpha/(2\pi) = -216.0$ MHz, with $\kappa = 0.00225$ MHz^{-1} for the corrected curve. The inset shows a ± 10 kHz center-frequency zoom; uncorrected echo points outside that window are clipped. These finite-pulse positions are operational spectral centers rather than instantaneous dressed-state shifts.

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